

# Whole-Body Vibration and the Prevention and Treatment of Delayed-Onset Muscle Soreness

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**Context:** Numerous recovery strategies have been used in an attempt to minimize the symptoms of delayed-onset muscle soreness (DOMS). Whole-body vibration (WBV) has been suggested as a viable warm-up for athletes. However, scientific evidence to support the protective effects of WBV training (WBVT) on muscle damage is lacking. **Objective:** To investigate the acute effect of WBVT applied before eccentric exercise in the prevention of DOMS. **Design:** Randomized controlled trial. **Setting:** University laboratory. **Patients or Other Participants:** A total of 32 healthy, untrained volunteers were randomly assigned to either the WBVT ( $n = 15$ ) or control ( $n = 17$ ) group. **Intervention(s):** Volunteers performed 6 sets of 10 maximal isokinetic ( $60^\circ/\text{s}$ ) eccentric contractions of the dominant-limb knee extensors on a dynamometer. In the WBVT group, the training was applied using a vibratory platform (35 Hz, 5 mm peak to peak) with  $100^\circ$  of knee flexion for 60 seconds before eccentric exercise. No vibration was applied in the control group. **Main Outcome Measure(s):** Muscle soreness, thigh circumference, and pressure pain threshold were recorded at baseline and at 1, 2, 3, 4, 7, and 14 days postexercise. Maximal voluntary isometric and isokinetic knee extensor strength were assessed at baseline, immediately after exercise, and at 1, 2, 7, and 14 days postexercise. Serum creatine kinase was measured at baseline and at 1, 2, and 7 days postexercise. **Results:** The WBVT group showed a reduction in DOMS symptoms in the form of less maximal isometric and isokinetic voluntary strength loss, lower creatine kinase levels, and less pressure pain threshold and muscle soreness ( $P < .05$ ) compared with the control group. However, no effect on thigh circumference was evident ( $P < .05$ ). **Conclusions:** Administered before eccentric exercise, WBVT may reduce DOMS via muscle function improvement. Further investigation should be undertaken to ascertain the effectiveness of WBVT in attenuating DOMS in athletes.